Q1 - 2024 (01 Feb Shift 1)

The value of the integral $\int_0^{\frac{\pi}{4}} \frac{x d x}{\sin ^4(2 x)+\cos ^4(2 x)}$ equals :

(1) $\frac{\sqrt{2} \pi^2}{8}$
(2) $\frac{\sqrt{2} \pi^2}{16}$
(3) $\frac{\sqrt{2} \pi^2}{32}$
(4) $\frac{\sqrt{2} \pi^2}{64}$

Q2 - 2024 (01 Feb Shift 1)

If $\int_{-\pi / 2}^{\pi / 2} \frac{8 \sqrt{2} \cos x d x}{\left(1+e^{\sin x}\right)\left(1+\sin ^4 x\right)}=\alpha \pi+\beta \log _e(3+2 \sqrt{2})$, where α, β are integers, then $\alpha^2+\beta^2$ equals

Q3 - 2024 (01 Feb Shift 2)

The value of $\int_0^1\left(2 x^3-3 x^2-x+1\right)^{\frac{1}{3}} d x$ is equal to:

(1) 0

(2) 1

(3) 2

(4) -1

Q4 - 2024 (01 Feb Shift 2)

If $\int_0^{\frac{\pi}{3}} \cos ^4 x d x=a \pi+b \sqrt{3}$, where a and b are rational numbers, then $9 a+8 b$ is equal to :

(1) 2

(2) 1

(3) 3

(4) $\frac{3}{2}$

Q5 - 2024 (01 Feb Shift 2)

Let $f:(0, \infty) \rightarrow R$ and $F(x)=\int_0^x t f(t) d t$. If $F\left(x^2\right)=\mathrm{x}^4+\mathrm{x}^5$, then $\sum_{\mathrm{r}=1}^{12} \mathrm{f}\left(\mathrm{r}^2\right)$ is equal to :

Q6 - 2024 (27 Jan Shift 1)

If (a, b) be the orthocentre of the triangle whose vertices are $(1,2),(2,3)$ and $(3,1)$, and $I_1=\int_a^b x \sin \left(4 x-x^2\right) d x, I_2=\int_a^b \sin \left(4 x-x^2\right) d x$, then $36 \frac{I_1}{I_2}$ is equal to :

(1) 72

(2) 88

(3) 80

(4) 66

Q7 - 2024 (27 Jan Shift 1)

If $\int_0^1 \frac{1}{\sqrt{3+x}+\sqrt{1+x}} d x=a+b \sqrt{2}+c \sqrt{3}$, where a, b, c are rational numbers, then $2 a+3 b-4 c$ is equal to :

(1) 4

(2) 10

(3) 7

(4) 8

Q8 - 2024 (27 Jan Shift 2)

For $0<\mathrm{a}<1$, the value of the integral $\int_0^\pi \frac{d x}{1-2 a \cos x+a^2}$ is :

(1) $\frac{\pi^2}{\pi+a^2}$

(2) $\frac{\pi^2}{\pi-a^2}$

(3) $\frac{\pi}{1-a^2}$

(4) $\frac{\pi}{1+a^2}$

Q9 - 2024 (27 Jan Shift 2)

Let $f(x)=\int_0^x g(t) \log_e\left(\frac{1-t}{1+t}\right) d t$, where g is a continuous odd function.

If $\int_{-\pi / 2}^{\pi / 2}\left(f(x)+\frac{x^2 \cos x}{1+e^x}\right) d x=\left(\frac{\pi}{\alpha}\right)^2-\alpha$, then α is equal to

Q10 - 2024 (29 Jan Shift 1)

If the value of the integral

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{x^2\cos x}{1+\pi^x}+\frac{1+\sin^2 x}{1+e^{\sin x^{2033}}}\right)dx=\frac{\pi}{4}(\pi+a)-2,$

then the value of a is

(1) 3

(2) $-\frac{3}{2}$

(3) 2

(4) $\frac{3}{2}$

Q11 - 2024 (29 Jan Shift 2)

If $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\sqrt{1-\sin 2x}dx=\alpha+\beta\sqrt{2}+\gamma\sqrt{3}$, where α,β and γ are rational numbers, then $3\alpha+4\beta-\gamma$ is equal to______

Q12 - 2024 (30 Jan Shift 1)

The value of $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n^3}{(n^2+k^2)(n^2+3k^2)}$ is:

(1) $\frac{(2\sqrt{3}+3)\pi}{24}$

(2) $\frac{13\pi}{8(4\sqrt{3}+3)}$

(3) $\frac{13(2\sqrt{3}-3)\pi}{8}$

(4) $\frac{\pi}{8(2\sqrt{3}+3)}$

Q13 - 2024 (30 Jan Shift 1)

The value $9\int_0^9\left[\sqrt{\frac{10x}{x+1}}\right]$ dx, where $[t]$ denotes the greatest integer less than or equal to t, is______

Q14 - 2024 (30 Jan Shift 2)

Let $y=f(x)$ be a thrice differentiable function in $(-5,5)$. Let the tangents to the curve $y=f(x)$ at $(1,\mathrm{f}(1))$ and $(3,\mathrm{f}(3))$ make angles $\frac{\pi}{6}$ and $\frac{\pi}{4}$, respectively with positive x-axis. If

$27\int_1^3\left(\left(f'(t)\right)^2+1\right)f''(t)dt=\alpha+\beta\sqrt{3}$ where α, β are integers, then the value of $\alpha+\beta$ equals

(1) -14

(2) 26

(3) -16

(4) 36

Q15 - 2024 (30 Jan Shift 2)

Let $f: R \rightarrow R$ be defined $f(x)=a e^{2 x}+b e^x+c x$. If $f(0)=-1, f'\left(\log _e 2\right)=21$ and $\int_0^{\log _4 4}(f(x)-c x) d x=\frac{39}{2}$, then the value of $|a+b+c|$ equals:

(1) 16

(2) 10

(3) 12

(4) 8

Q16 - 2024 (31 Jan Shift 1)

If the integral $525 \int_0^{\frac{\pi}{2}} \sin 2 x \cos ^{\frac{11}{2}} x\left(1+\cos ^{\frac{5}{2}} x\right)^{\frac{1}{2}} d x$ is equal to $(n \sqrt{2}-64)$, then n is equal to_______.

Q17 - 2024 (31 Jan Shift 1)

Let $S=(-1, \infty)$ and $f: S \rightarrow \mathbb{R}$ be defined as

$f(x)=\int_{-1}^x\left(e^t-1\right)^{11}(2 t-1)^5(t-2)^7(t-3)^{12}(2 t-10)^{61} d t$

Let $p=$ Sum of square of the values of x, where $f(x)$ attains local maxima on S. and $q=$ Sum of the values of x, where $f(x)$ attains local minima on S. Then, the value of $p^2+2 q$ is_______.

Q18 - 2024 (31 Jan Shift 1)

Let f : $\mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x)=\frac{4^x}{4^x+2}$ and

$$M=\int_{f(a)}^{f(1-a)} x \sin ^4(x(1-x)) d x$$

$$N=\int_{f(a)}^{f(1-a)} \sin ^4(x(1-x)) d x ; a \neq \frac{1}{2} . \text { If }$$

$\alpha \mathrm{M}=\beta \mathrm{N}, \alpha, \beta \in \mathbb{N}$, then the least value of $\alpha^2+\beta^2$ is equal to _______.

Q19 - 2024 (31 Jan Shift 2)

Let $f, g : (0, \infty) \rightarrow R$ be two functions defined by $f(x) = \int_{-x}^{x} \left(|t| - t^2\right) e^{-t^2} dt$ and $g(x) = \int_0^{x^2} t^{1/2} e^{-t} dt$. Then the value of $\left(f\left(\sqrt{\log_e 9}\right) + g\left(\sqrt{\log_e 9}\right)\right)$ is equal to

(1) 6

(2) 9

(3) 8

(4) 10

Q20 - 2024 (31 Jan Shift 2)

$\left|\frac{120}{\pi^3} \int_0^{\pi} \frac{x^2 \sin x \cos x}{\sin^4 x + \cos^4 x} dx\right|$ is equal to ________.

Answer Key

Q1 (3)	**Q2** (8)	**Q3** (1)	**Q4** (1)
Q5 (219)	**Q6** (1)	**Q7** (4)	**Q8** (3)
Q9 (2)	**Q10** (1)	**Q11** (6)	**Q12** (2)
Q13 (155)	**Q14** (2)	**Q15** (4)	**Q16** (176)
Q17 (27)	**Q18** (5)	**Q19** (3)	**Q20** (15)

Q1

$\int_0^{\frac{\pi}{4}} \frac{x dx}{\sin^4(2x)+\cos^4(2x)}$

Let $2x = t$ then $dx = \frac{1}{2} dt$

$$I = \frac{1}{4} \int_0^{\frac{\pi}{2}} \frac{t dt}{\sin^4 t + \cos^4 t}$$

$$I = \frac{1}{4} \int_0^{\frac{\pi}{2}} \frac{\left(\frac{\pi}{2} - t\right) dt}{\sin^4\left(\frac{\pi}{2} - t\right) + \cos^4\left(\frac{\pi}{2} - t\right)}$$

$$I = \frac{1}{4} \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2} dt}{\sin^4 t + \cos^4 t} - I$$

$$2I = \frac{\pi}{8} \int_0^{\frac{\pi}{2}} \frac{dt}{\sin^4 t + \cos^4 t}$$

$$2I = \frac{\pi}{8} \int_0^{\frac{\pi}{2}} \frac{\sec^4 t dt}{\tan^4 t + 1}$$

Let $\tan t = \mathrm{y}$ then $\sec^2 t dt = \mathrm{dy}$

$$2I = \frac{\pi}{8} \int_0^{\infty} \frac{\left(1 + y^2\right) dy}{1 + y^4}$$

$$= \frac{\pi}{16} \int_0^{\infty} \frac{1 + \frac{1}{\mathrm{y}^2}}{\mathrm{y}^2 + \frac{1}{\mathrm{y}^2}} \mathrm{dy}$$

Put $y - \frac{1}{y} = p$

$$I = \frac{\pi}{16} \int_{-\infty}^{\infty} \frac{\mathrm{dp}}{\mathrm{p}^2 + (\sqrt{2})^2}$$

$$= \frac{\pi}{16\sqrt{2}} \left[\tan^{-1}\left(\frac{\mathrm{p}}{\sqrt{2}}\right)\right]_{-\infty}^{\infty}$$

$$I = \frac{\pi^2}{16\sqrt{2}}$$

Q2

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2} \cos x}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)} dx$$

Apply king

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{8\sqrt{2} \cos x \left(e^{\sin x}\right)}{\left(1+e^{\sin x}\right)\left(1+\sin^4 x\right)} dx$$

adding (1) & (2)

$$2I=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x}{1+\sin^4 x}dx$$

$$I=\int_0^{\frac{\pi}{2}}\frac{8\sqrt{2}\cos x}{1+\sin^4 x}dx$$

$\sin x=t$

$$I=\int_0^1\frac{8\sqrt{2}}{1+t^4}dx$$

$$I=4\sqrt{2}\int_0^1\left(\frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}}-\frac{1-\frac{1}{t^2}}{t^2+\frac{1}{t^2}}dt\right.$$

$$I=4\sqrt{2}\int_0^1\frac{\left(1+\frac{1}{t^2}\right)}{\left(t-\frac{1}{t}\right)^2+2}-\frac{1}{\left(t+\frac{1}{t}\right)^2-2}dt$$

Let $t-\frac{1}{t}=z \& t+\frac{1}{t}=k$

$$=4\sqrt{2}\left[\int_{-\infty}^0\frac{dz}{z^2+2}-\int_{\infty}^2\frac{dk}{k^2-2}\right]$$

$$=4\sqrt{2}\left[\frac{1}{\sqrt{2}}\tan^{-1}\frac{z}{\sqrt{2}}\right]_{-\infty}^0-\left[\frac{1}{2\sqrt{2}}\ln\left(\frac{k-\sqrt{2}}{k+\sqrt{2}}\right)\right]_{\infty}^2$$

$$=4\sqrt{2}\left[\frac{\pi}{2\sqrt{2}}-\frac{1}{2\sqrt{2}}\left[\ln\frac{2-\sqrt{2}}{2+\sqrt{2}}\right]\right]$$

$$=2\pi+2\ln(3+2\sqrt{2})$$

$\alpha=2$
$\beta=2$

Q3

$I=\int_0^1\left(2x^3-3x^2-x+1\right)^{\frac{1}{3}}dx$

Using $\int_0^{2a}f(x)dx$ where $f(2a-x)=-f(x)$ Here

$f(1-x)=f(x)$

$\therefore I=0$

Q4

$$\int_0^{\pi/3} \cos^4 x dx$$

$$= \int_0^{\pi/3} \left(\frac{1+\cos 2x}{2}\right)^2 dx$$

$$= \frac{1}{4} \int_0^{\pi/3} \left(1+2\cos 2x+\cos^2 2x\right) dx$$

$$= \frac{1}{4}\left[\int_0^{\pi/3} \mathrm{dx} + 2\int_0^{\pi/3} \cos 2\mathrm{x} dx + \int_0^{\pi/3} \frac{1+\cos 4\mathrm{x}}{2} \mathrm{dx}\right]$$

$$= \frac{1}{4}\left[\frac{\pi}{3} + (\sin 2\mathrm{x})_0^{\pi/3} + \frac{1}{2}\left(\frac{\pi}{3}\right) + \frac{1}{8}(\sin 4\mathrm{x})_0^{\pi/3}\right]$$

$$= \frac{1}{4}\left[\frac{\pi}{3} + (\sin 2\mathrm{x})_0^{\pi/3} + \frac{1}{2}\left(\frac{\pi}{3}\right) + \frac{1}{8}(\sin 4\mathrm{x})_0^{\pi/3}\right]$$

$$= \frac{1}{4}\left[\frac{\pi}{2} + \frac{\sqrt{3}}{2} + \frac{1}{8} \times \left(-\frac{\sqrt{3}}{2}\right)\right]$$

$$= \frac{\pi}{2} + \frac{7\sqrt{3}}{64}$$

$$\therefore \mathrm{a} = \frac{1}{8}; \mathrm{b} = \frac{7}{64}$$

$$\therefore 9\mathrm{a} + 8\mathrm{~b} = \frac{9}{8} + \frac{7}{8} = 2$$

Q5

$$\mathrm{F}(\mathrm{x}) = \int_0^{\mathrm{x}t} \mathrm{t} \cdot \mathrm{f}(\mathrm{t}) \mathrm{dt}$$

$$\text{Given} \quad \begin{array}{l} \mathrm{F}^1(\mathrm{x}) = \mathrm{xf}(\mathrm{x}) \\ \mathrm{F}\left(\mathrm{x}^2\right) = \mathrm{x}^4 + \mathrm{x}^5, \quad \text{let } \mathrm{x}^2 = \mathrm{t} \end{array}$$

$$F(t) = t^2 + t^{5/2}$$

$$F'(t) = 2t + 5/2t^{3/2}$$

$$t \cdot f(t) = 2t + 5/2t^{3/2}$$

$$f(t) = 2 + 5/2r^{1/2}$$

$$\sum_{r=1}^{12} f\left(r^2\right) = \sum_{r=1}^{12} 2 + \frac{5}{2} r$$

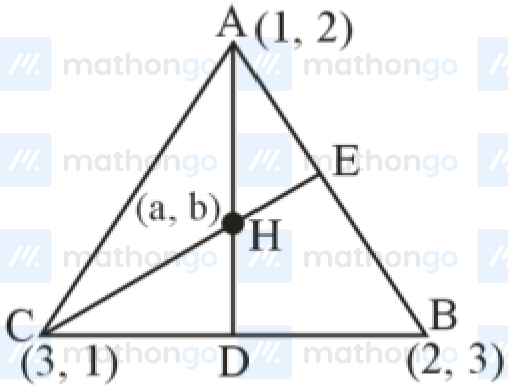
$$= 24 + 5/2\left[\frac{12(13)}{2}\right]$$

$$= 219$$

Q6

Equation of CE

$y-1=-(x-3)$
$x+y=4$



orthocentre lies on the line $x+y=4$

so, $a+b=4$

$I_1=\int_a^b x \sin (x(4-x)) d x$ ...(i)

Using king rule

$I_1=\int_a^b(4-x) \sin (x(4-x)) d x$ ...(ii)

(i) + (ii)

$2 I_1=\int_a^b 4 \sin (x(4-x)) d x$
$2\mathrm{I}_1=4\mathrm{I}_2$
$\mathrm{I}_1=2\mathrm{I}_2$
$\frac{\mathrm{I}_1}{\mathrm{I}_2}=2$
$\frac{36\mathrm{I}_1}{\mathrm{I}_2}=72$

Q7

$$\int_0^1 \frac{1}{\sqrt{3+x}+\sqrt{1+x}} dx=\int_0^1 \frac{\sqrt{3+x}-\sqrt{1+x}}{(3+x)-(1+x)} dx$$

$$\frac{1}{2}\left[\int_0^1 \sqrt{3+x} dx-\int_0^1(\sqrt{1+x}) dx\right]$$

$$\frac{1}{2}\left[2 \frac{(3+x)^{\frac{3}{2}}}{3}-\frac{2(1+x)^{\frac{3}{2}}}{3}\right]_0^1$$

$$\frac{1}{2}\left[\frac{2}{3}(8-3 \sqrt{3})-\frac{2}{3}\left(2^{\frac{3}{2}}-1\right)\right]$$

$$\frac{1}{3}[8-3 \sqrt{3}-2 \sqrt{2}+1]$$

$$=3-\sqrt{3}-\frac{2}{3} \sqrt{2}=a+b \sqrt{2}+c \sqrt{3}$$

$$a=3, b=-\frac{2}{3}, c=-1$$

$$2 a+3 b-4 c=6-2+4=8$$

Q8

$$I=\int_0^\pi \frac{d x}{1-2 a \cos x+a^2} ; 0<a<1$$

$$I=\int_0^\pi \frac{d x}{1+2 a \cos x+a^2}$$

$$2 \mathrm{I}=2 \int_0^{\pi / 2} \frac{2\left(1+\mathrm{a}^2\right)}{\left(1+\mathrm{a}^2\right)^2-4 \mathrm{a}^2 \cos ^2 \mathrm{x}} \mathrm{dx}$$

$$\Rightarrow \mathrm{I}=\int_0^{\pi / 2} \frac{2\left(1+\mathrm{a}^2\right) \cdot \sec ^2 \mathrm{x}}{\left(1+\mathrm{a}^2\right)^2 \cdot \sec ^2 \mathrm{x}-4 \mathrm{a}^2} \mathrm{dx}$$

$$\Rightarrow \mathrm{I}=\int_0^{\pi / 2} \frac{2 \cdot\left(1+\mathrm{a}^2\right) \cdot \sec ^2 \mathrm{x}}{\left(1+\mathrm{a}^2\right)^2 \cdot \tan ^2 \mathrm{x}+\left(1-\mathrm{a}^2\right)^2} \mathrm{dx}$$

$$\Rightarrow \mathrm{I}=\int_0^{\pi / 2} \frac{\frac{2 \cdot \sec ^2 \mathrm{x}}{1+\mathrm{a}^2} \cdot \mathrm{dx}}{\tan ^2 \mathrm{x}+\left(\frac{1-\mathrm{a}^2}{1+\mathrm{a}^2}\right)^2}$$

$$\Rightarrow \mathrm{I}=\frac{2}{\left(1-\mathrm{a}^2\right)}\left[\frac{\pi}{2}-0\right]$$

$$I=\frac{\pi}{1-\mathrm{a}^2}$$

Q9

$f(x)=\int_0^x g(t)\ln\left(\frac{1-t}{1+t}\right)dt$

$f(-x)=\int_0^{-x} g(t)\ln\left(\frac{1-t}{1+t}\right)dt$

$f(-x)=-\int_0^x g(-y)\ln\left(\frac{1+y}{1-y}\right)dy$

$=-\int_0^x g(y)\ln\left(\frac{1-y}{1+y}\right)dy$ (g is odd)

$f(-x)=-f(x) \Rightarrow f$ is also odd

Now,

$$I=\int_{-\pi/2}^{\pi/2}\left(f(x)+\frac{x^2\cos x}{1+e^x}\right)dx\ldots(1)$$

$$I=\int_{-\pi/2}^{\pi/2}\left(f(-x)+\frac{x^2e^x\cos x}{1+e^x}\right)dx\ldots(2)$$

$$2I=\int_{-\pi/2}^{\pi/2}x^2\cos x dx=2\int_0^{\pi/2}x^2\cos x dx$$

$$I=\left(x^2\sin x\right)_0^{\pi/2}-\int_0^{\pi/2}2x\sin x dx$$

$$=\frac{\pi^2}{4}-2\left(-x\cos x+\int\cos x dx\right)_0^{\pi/2}$$

$$=\frac{\pi^2}{4}-2(0+1)=\frac{\pi^2}{4}-2\Rightarrow\left(\frac{\pi}{2}\right)^2-2$$

$\therefore \alpha=2$

Q10

$$I=\int_{-\pi/2}^{\pi/2}\left(\frac{x^2\cos x}{1+\pi^x}+\frac{1+\sin^2 x}{1+e^{\sin x^{2023}}}\right)dx$$

$$I=\int_{-\pi/2}^{\pi/2}\left(\frac{x^2\cos x}{1+\pi^{-x}}+\frac{1+\sin^2 x}{1+e^{\sin(-x)^{2023}}}\right)dx$$

On Adding, we get

$2I=\int_{-\pi/2}^{\pi/2}\left(x^2\cos x+1+\sin^2 x\right)dx$

On solving

$$I=\frac{\pi^2}{4}+\frac{3\pi}{4}-2$$

$a=3$

Q11

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{1 - \sin 2x} dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin x - \cos x \Big| \; dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (\cos x - \sin x) dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} (\sin x - \cos x) dx$$

$$= -1 + 2\sqrt{2} - \sqrt{3}$$

$$= \alpha + \beta\sqrt{2} + \gamma\sqrt{3}$$

$$\alpha = -1, \beta = 2, \gamma = -1$$

$$3\alpha + 4\beta - \gamma = 6$$

Q12

$$\lim_{n \to \infty} \sum_{k=1}^{n} \frac{n^3}{n^4 \left(1 + \frac{k^2}{n^2}\right)\left(1 + \frac{3k^2}{n^2}\right)}$$

$$= \lim_{n \to \infty} \frac{1}{n} \sum_{k=1}^{n} \frac{n^3}{\left(1 + \frac{k^2}{n^2}\right)\left(1 + \frac{3k^2}{n^2}\right)}$$

$$= \int_0^1 \frac{dx}{3\left(1 + x^2\right)\left(\frac{1}{3} + x^2\right)}$$

$$= \int_0^1 \frac{1}{3} \times \frac{3}{2} \frac{\left(x^2 + 1\right) - \left(x^2 + \frac{1}{3}\right)}{\left(1 + x^2\right)\left(x^2 + \frac{1}{3}\right)} dx$$

$$= \frac{1}{2} \int_0^1 \left[\frac{1}{x^2 + \left(\frac{1}{\sqrt{3}}\right)^2} - \frac{1}{1 + x^2}\right] dx$$

$$= \frac{1}{2} \left[\sqrt{3} \tan^{-1}(\sqrt{3}x)\right]_0^1 - \frac{1}{2} \left(\tan^{-1} x\right)_0^1$$

$$= \frac{\sqrt{3}}{2}\left(\frac{\pi}{3}\right) - \frac{1}{2}\left(\frac{\pi}{4}\right) = \frac{\pi}{2\sqrt{3}} - \frac{\pi}{8}$$

$$= \frac{13\pi}{8 \cdot (4\sqrt{3} + 3)}$$

Q13

$\frac{10x}{x+1} = 1 \quad \Rightarrow x = \frac{1}{9}$

$\frac{10x}{x+1} = 4 \quad \Rightarrow x = \frac{2}{3}$

$\frac{10x}{x+1} = 9 \quad \Rightarrow x = 9$

$$I = 9\left(\int_0^{1/9} 0dx + \int_{1/9}^{2/3} 1dx + \int_{2/3}^{9} 2dx\right)$$

$= 155$

Q14

$$y = f(x) \Rightarrow \frac{dy}{dx} = f'(x)$$

$$\left.\frac{dy}{dx}\right)_{(l,(1))} = f'(l) = \tan\frac{\pi}{6} = \frac{1}{\sqrt{3}} \Rightarrow f'(l) = \frac{1}{\sqrt{3}}$$

$$\left.\frac{dy}{dx}\right)_{(3,r(3))} = f'(3) = \tan\frac{\pi}{4} = 1 \Rightarrow f'(3) = 1$$

$$27\int_1^3 \left(\left(f'(t)\right)^2 + 1\right) f''(t)dt = \alpha + \beta\sqrt{3}$$

$$I = \int_1^3 \left(\left(f'(t)\right)^2 + 1\right) f''(t)dt$$

$$f(t) = z \Rightarrow f'(t)dt = dz$$

$$z = f(3) = 1$$

$$z = f(1) = \frac{1}{\sqrt{3}}$$

$$I = \int_{1/\sqrt{3}}^{1} \left(z^2 + 1\right) dz = \left(\frac{z^3}{3} + z\right)_{1/\sqrt{5}}^{1}$$

$$= \left(\frac{1}{3} + 1\right) - \left(\frac{1}{3}\cdot\frac{1}{3\sqrt{3}} + \frac{1}{\sqrt{3}}\right)$$

$$= \frac{4}{3} - \frac{10}{9\sqrt{3}} = \frac{4}{3} - \frac{10}{27}\sqrt{3}$$

$$\alpha + \beta\sqrt{3} = 27\left(\frac{4}{3} - \frac{10}{27}\sqrt{3}\right) = 36 - 10\sqrt{3}$$

$$\alpha = 36, \beta = -10$$

$$\alpha + \beta = 36 - 10 = 26$$

Q15

$f(x)=ae^{2x}+be^{x}+cx \quad f(0)=-1$

$\mathrm{a}+\mathrm{b}=-1$

$f'(x)=2ae^{2x}+be^{x}+c \quad f'(\ln 2)=21$

$8a+2b+c=21$

$$\int_0^{\ln 4}\left(ae^{2x}+be^{x}\right)dx=\frac{39}{2}$$

$$\left[\frac{\mathrm{ae}^{2\mathrm{x}}}{2}+\mathrm{be}^{\mathrm{x}}\right]_0^{\ln 4}=\frac{39}{2}\Rightarrow 8\mathrm{a}+4\,\mathrm{b}-\frac{\mathrm{a}}{2}-\mathrm{b}=\frac{39}{2}$$

$15a+6b=39$

$15a-6a-6=39$

$9a=45\Rightarrow a=5$

$b=-6$

$c=21-40+12=-7$

$a+b+c-8$

$|a+b+c|=8$

Q16

$I=\int_0^{\frac{\pi}{2}}\sin 2x\cdot(\cos x)^{\frac{11}{2}}\left(1+(\cos x)^{\frac{5}{2}}\right)^{\frac{1}{2}}dx$

Put $\cos x=t^2\Rightarrow \sin x dx=-2tdt$

$$\therefore \mathrm{I}=4\int_0^1 \mathrm{t}^2\cdot\mathrm{t}^{11}\sqrt{\left(1+\mathrm{t}^5\right)}(\mathrm{t})\mathrm{dt}$$

$$\mathrm{I}=4\int_0^1\mathrm{t}^{14}\sqrt{1+\mathrm{t}^5}\mathrm{dt}$$

Put $1+\mathrm{t}^5==\mathrm{k}^2$

$\Rightarrow 5t^4\mathrm{dt}=2\mathrm{kdk}$

$$\therefore \mathrm{I}=4\cdot\int_1^{\sqrt{2}}\left(\mathrm{k}^2-1\right)^2\cdot\mathrm{k}\frac{2\mathrm{k}}{5}\mathrm{dk}$$

$$\mathrm{I}=\frac{8}{5}\int_1^{\sqrt{2}}\mathrm{k}^6-2\mathrm{k}^4+\mathrm{k}^2\mathrm{dk}$$

$\mathrm{I}=\frac{8}{5}\left[\frac{\mathrm{k}^7}{7}-\frac{2\mathrm{k}^5}{5}+\frac{\mathrm{k}^3}{3}\right]_1^{\sqrt{2}}$

$\mathrm{I}=\frac{8}{5}\left[\frac{8\sqrt{2}}{7}-\frac{8\sqrt{2}}{5}+\frac{2\sqrt{2}}{3}-\frac{1}{7}+\frac{2}{5}-\frac{1}{3}\right]$

$\mathrm{I}=\frac{8}{5}\left[\frac{22\sqrt{2}}{105}-\frac{8}{105}\right]$

$\therefore 525\cdot I=176\sqrt{2}-64$

Q17

$f'(x)=(e^x-1)^{11}(2x-1)^5(x-2)^7(x-3)^{12}(2x-10)^{61}$

Local minima at $x=\frac{1}{2}, x=5$

Local maxima at $x=0, x=2$

$\therefore p=0+4=4, q=\frac{1}{2}+5=\frac{11}{2}$

Then $p^2+2q=16+11=27$

Q18

$f(a)+f(1-a)=1$

$M=\int_{f(a)}^{f(1-a)}(1-x)\cdot\sin^4 x(1-x)dx$

$M=N-M \quad 2M=N$

$\alpha=2;\beta=1;$

Ans. 5

Q19

$f(x)=\int_{-x}^{x}\left(|t|-t^2\right)e^{-t^2}dt$

$\Rightarrow f'(x)=2\cdot\left(|x|-x^2\right)e^{-x^2}\ldots\ldots\ldots\ldots(1)$

$g(x)=\int_0^{x^2}t^{\frac{1}{2}}e^{-t}dt$

$g'(x)=xe^{-x^2}(2x)-0$

$f'(x)+g'(x)=2xe^{-x^2}-2x^2e^{-x^2}+2x^2e^{-x^2}$

Integrating both sides w.r.t. x

$f(x)+g(x)=\int_0^{\alpha}2xe^{-x^2}dx$

$x^2=t$

$\Rightarrow\int_0^{\sqrt{\alpha}}e^{-t}dt=\left[-e^{-t}\right]_0^{\sqrt{\alpha}}$

$=-e^{(\log_0(9)^{-1})+1}$

$\Rightarrow 9(f(x)+g(x))=\left(1-\frac{1}{9}\right)9=8$

Q20

Solutions **MathonGo**

$$\int_0^\pi \frac{x^2 \sin x \cdot \cos x}{\sin^4 x+\cos^4 x} dx$$

$$=\int_0^{\frac{\pi}{2}} \frac{\sin x \cdot \cos x}{\sin^4 x+\cos^4 x}\left(x^2-(\pi-x)^2\right) dx$$

$$=\int_0^{\frac{\pi}{2}} \frac{\sin x \cdot \cos x\left(2 \pi x-\pi^2\right)}{\sin^4 x+\cos^4 x}$$

$$=2 \pi \int_0^{\frac{\pi}{2}} \frac{x \sin x \cos x}{\sin^4 x+\cos 4 x} dx-\pi^2 \int_0^{\frac{\pi}{2}} \frac{\sin x \cos x}{\sin^4 x+\cos^4 x} dx$$

$$=2 \pi \cdot \frac{\pi}{4} \int_0^{\frac{\pi}{2}} \frac{\sin x \cos^4 x}{\sin^4 x+\cos^4 x} dx-\pi^2 \int_0^{\frac{\pi}{2}} \frac{\sin x \cos^4 x}{\sin^4 x+\cos^4 x} dx$$

$$=-\frac{\pi^2}{2} \int_0^{\frac{\pi}{2}} \frac{\sin x \cos x}{\sin^4 x+\cos^4 x} dx$$

$$=-\frac{\pi^2}{2} \int_0^{\frac{\pi}{2}} \frac{\sin x \cos x d x}{1-2 \sin^2 x \times \cos^2 x}$$

$$=-\frac{\pi^2}{2} \int_0^{\frac{\pi}{2}} \frac{\sin 2 x}{2-\sin^2 2 x} dx$$

$$=-\frac{\pi^2}{2} \int_0^{\frac{\pi}{2}} \frac{\sin 2 x}{1+\cos^2 2 x} dx$$

Let $\cos 2\mathrm{x}=\mathrm{t}$